

LAB 05: GUI PROGRAMMING

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Ngày nộp bài: 02/06/2026

A. GUI PROJECT

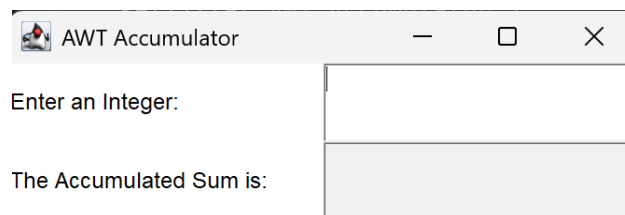
1. SWINGS

1.1. AWTAccumulator

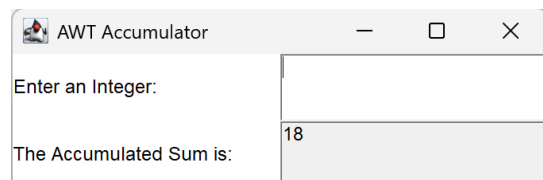
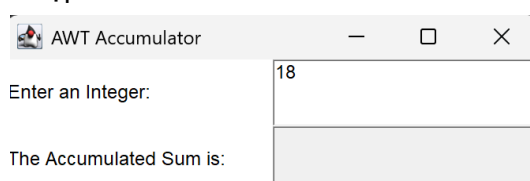
Giao diện cho phép nhập số và cộng dồn kết quả.

Người dùng nhập số và nhấn ENTER để tính tổng.

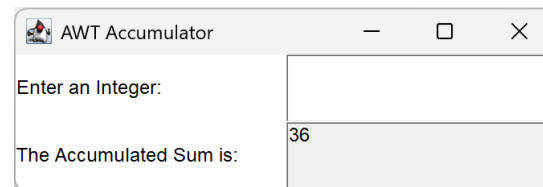
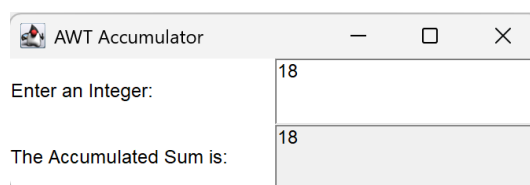
- Giao diện



- Nhập vào 1 số rồi ấn ENTER



- Nhập tiếp 1 số và ENTER

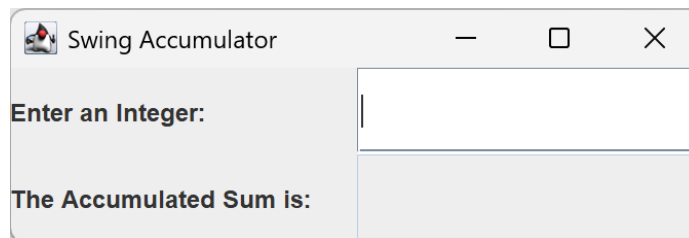


1.2. SwingAccumulator

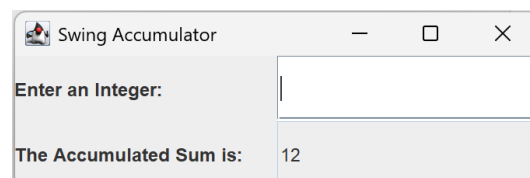
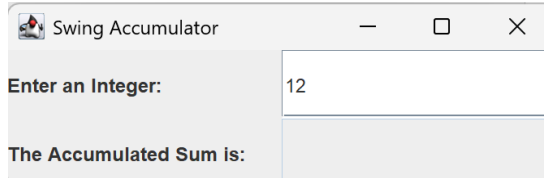
Tương tự AWTAccumulator nhưng sử dụng Swing.

Cho phép nhập nhiều số liên tiếp và cộng dồn.

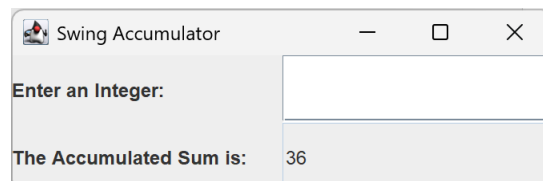
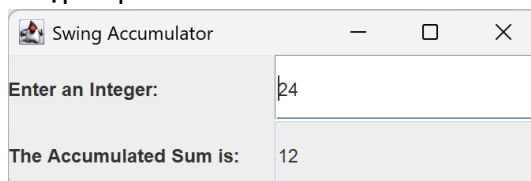
- Giao diện



- Nhập vào 1 số rồi ấn ENTER



- Nhập tiếp 1 số và ENTER

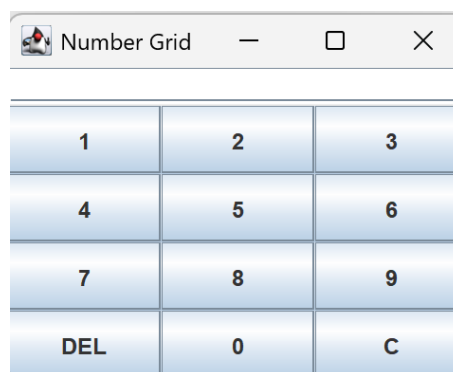


1.3. NumberGrid

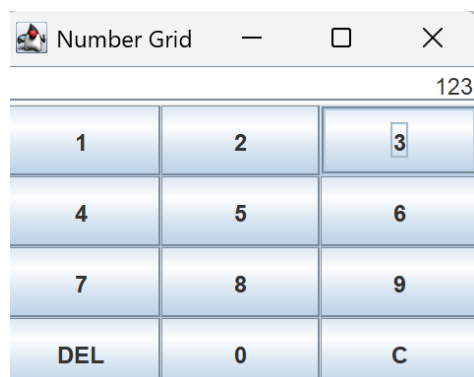
Hiển thị bảng số.

Có chức năng xóa (DEL) và reset (C).

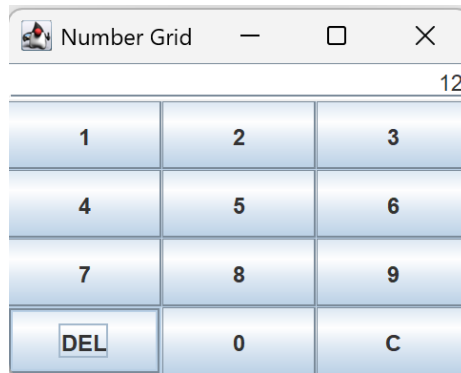
- Giao diện



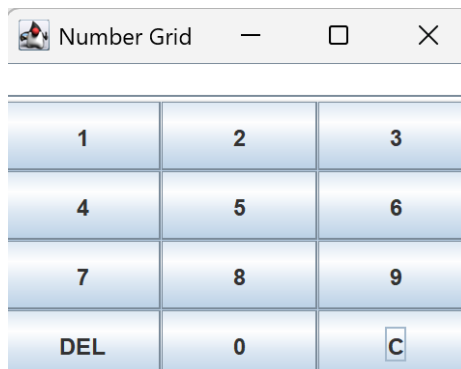
- Nhập số



- Ấn DEL



- Ấn C



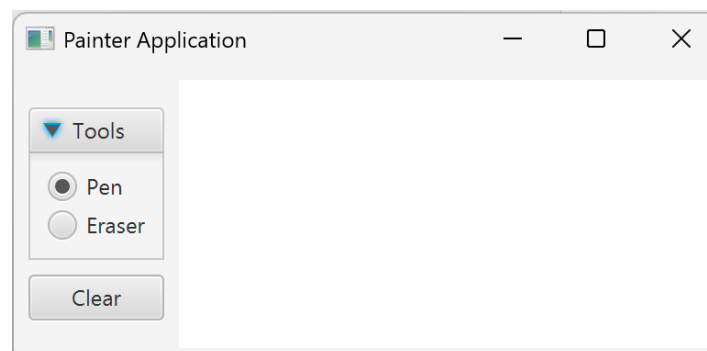
2. JAVAFX. PAINTER

Cho phép vẽ tự do bằng chuột (PEN).

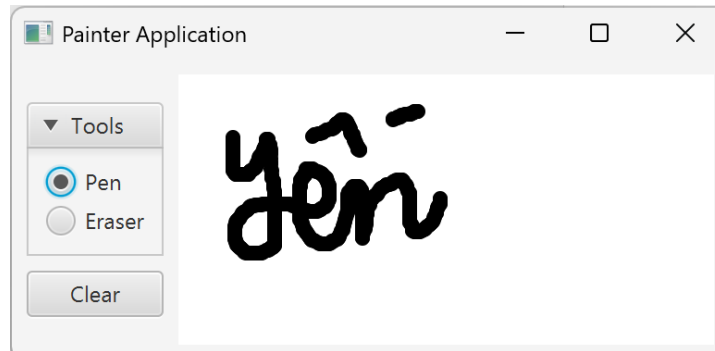
Có công cụ xóa (ERASER).

Có nút CLEAR để xóa toàn bộ bản vẽ.

- Giao diện



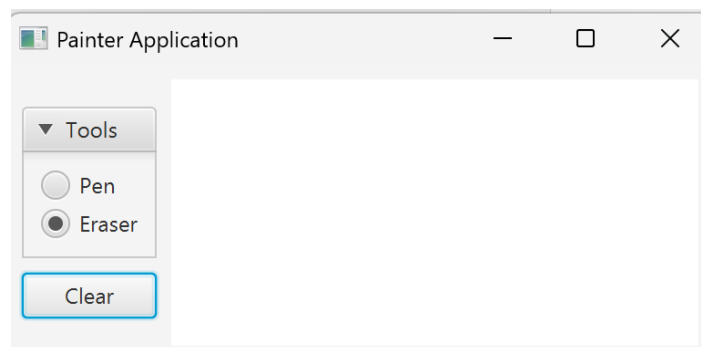
- Vẽ PEN



- Xóa ERASER



- Nhấn CLEAR

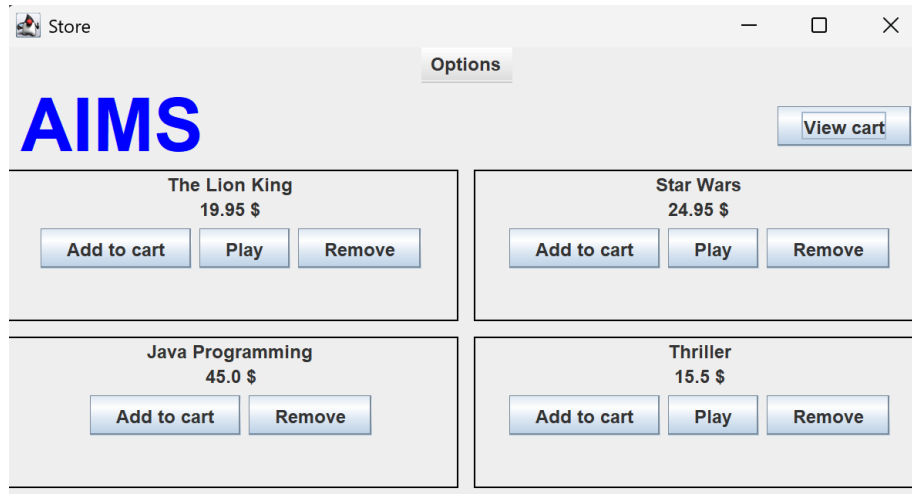


B. AIMS PROJECT

1. Giao diện Store

Hiển thị danh sách sản phẩm (Book, CD, DVD).

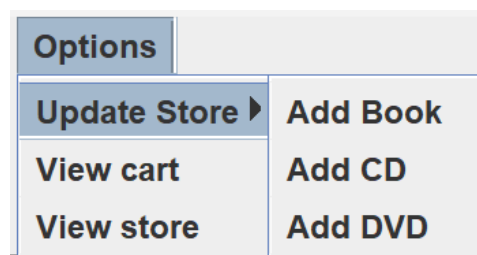
Cho phép thêm vào giỏ hàng.



2. Thanh OPTIONS

Menu điều hướng giữa Store và Cart.

Cho phép thêm Book, CD, DVD.



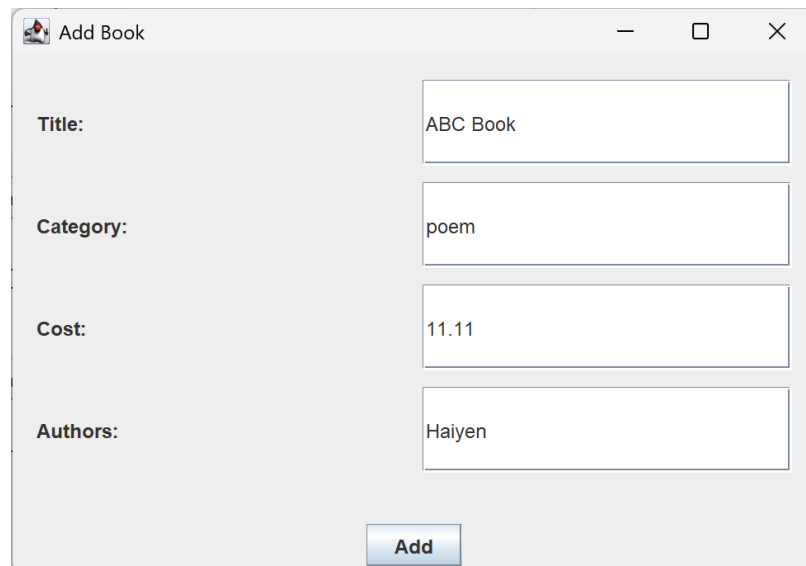
3. Add BOOK

Trong hệ thống, ID của mỗi sản phẩm được sinh tự động bằng biến static autoId.

Mỗi khi tạo một đối tượng Media mới, ID sẽ tăng dần 1 đơn vị và không nhập thủ công.

Cách làm này giúp đảm bảo:

- + Mỗi sản phẩm có ID duy nhất
- + ID tăng theo thứ tự thêm vào hệ thống
- + Hỗ trợ dễ dàng cho chức năng tìm kiếm theo ID



Add Book

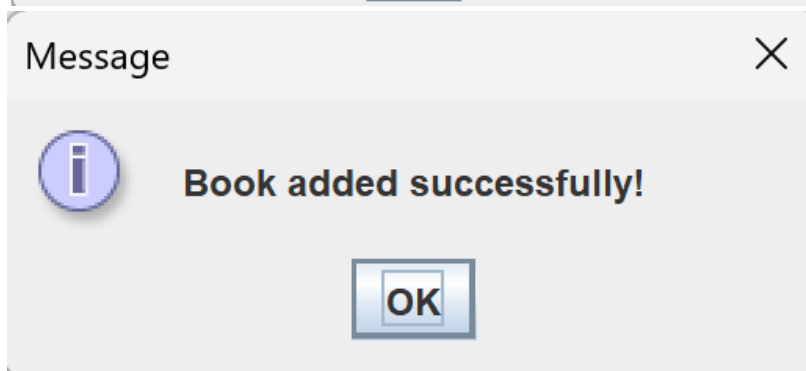
Title: ABC Book

Category: poem

Cost: 11.11

Authors: Haiyen

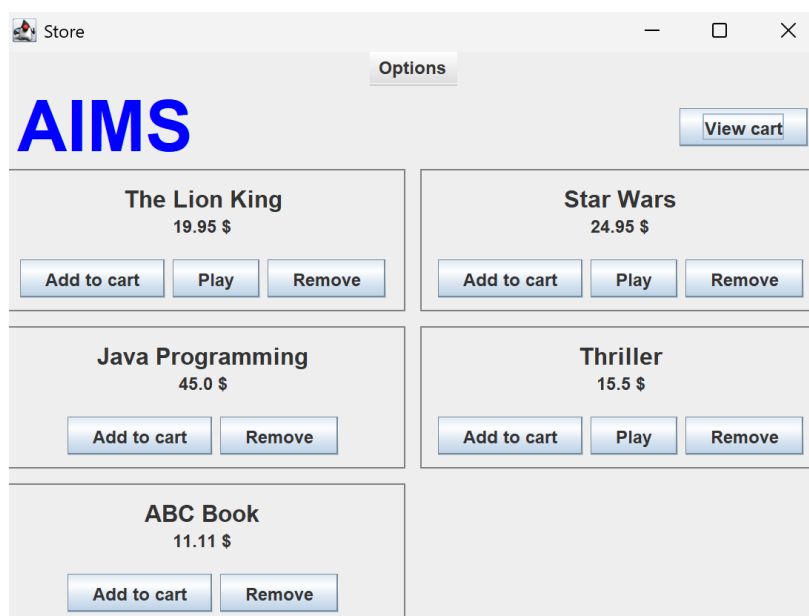
Add



Message

 **Book added successfully!**

OK



Store

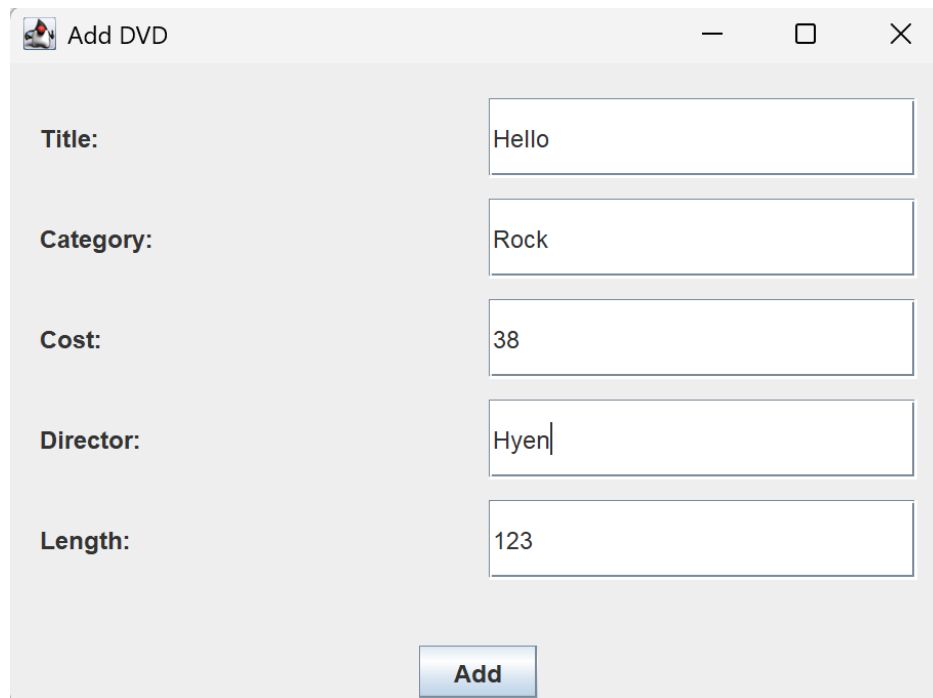
Options

AIMS

View cart

The Lion King 19.95 \$ Add to cart Play Remove	Star Wars 24.95 \$ Add to cart Play Remove
Java Programming 45.0 \$ Add to cart Remove	Thriller 15.5 \$ Add to cart Play Remove
ABC Book 11.11 \$ Add to cart Remove	

4. Add DVD

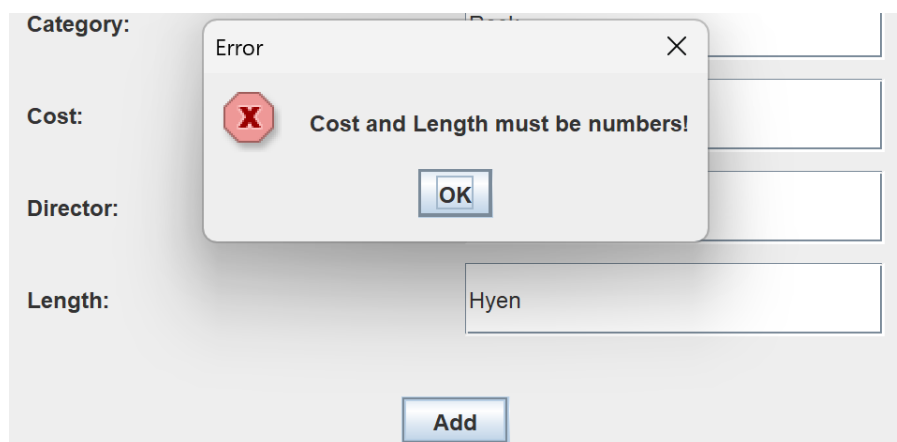


The screenshot shows a window titled "Add DVD" with a standard Windows title bar (minimize, maximize, close buttons). Inside the window, there is a form with five labels and corresponding text input fields:

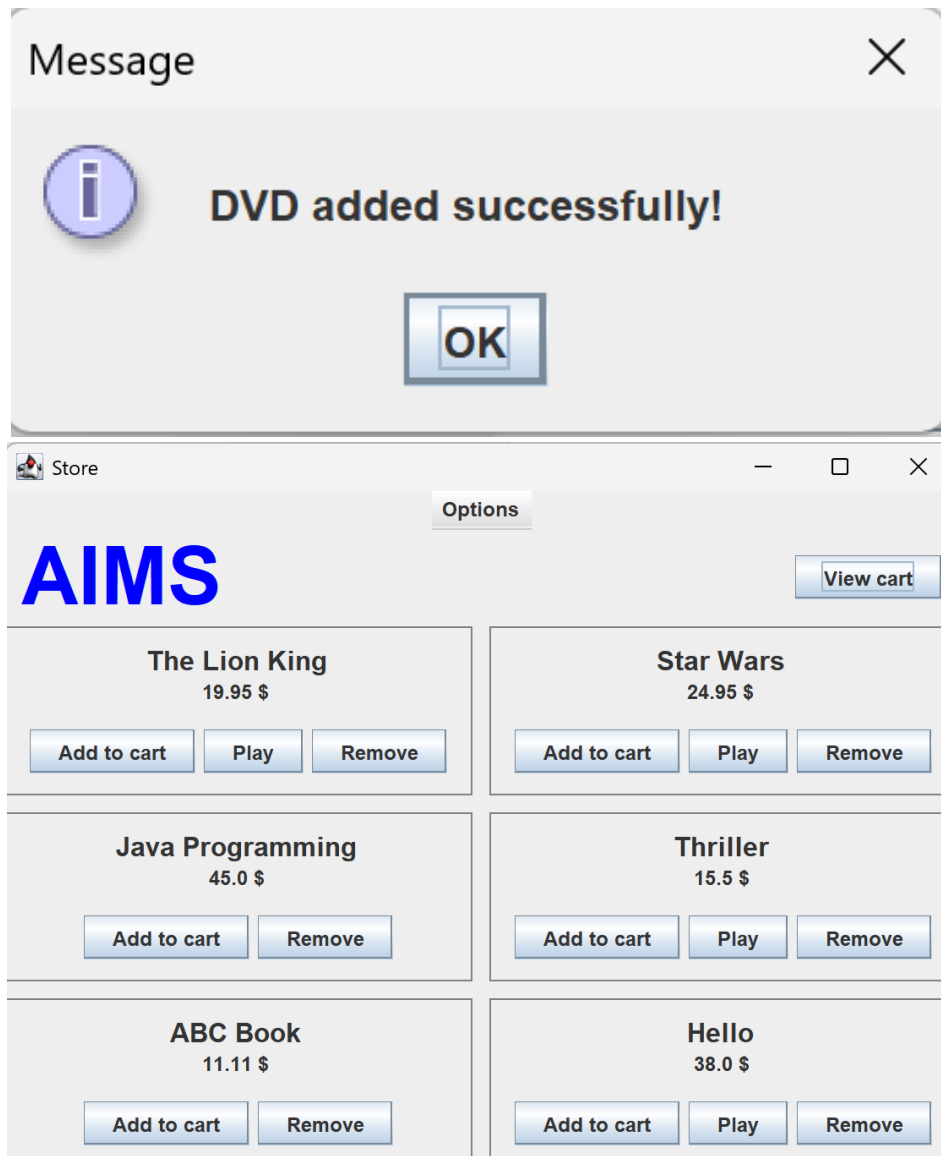
- Title:** Hello
- Category:** Rock
- Cost:** 38
- Director:** Hyen
- Length:** 123

At the bottom center of the form is a blue button labeled "Add".

- Nếu nhập Length là chữ (**KHÔNG ĐÚNG ĐỊNH DẠNG**):



This screenshot shows the same "Add DVD" window, but with an error dialog box overlaid in the center. The dialog box has a title bar that says "Error" and a close button (X). It contains a red octagonal icon with a white 'X' and the text "Cost and Length must be numbers!". There is an "OK" button at the bottom of the dialog. In the background, the "Length" field now contains the text "Hyen" instead of a number, and the "Add" button is still visible at the bottom.



5. Add CD

 Add CD

Title:

LaLaLa

Category:

Rock

Cost:

08.12

Artist:

Hy

Director:

Hy

Add

Message
 ✕


CD added successfully!

OK

 Store

Options



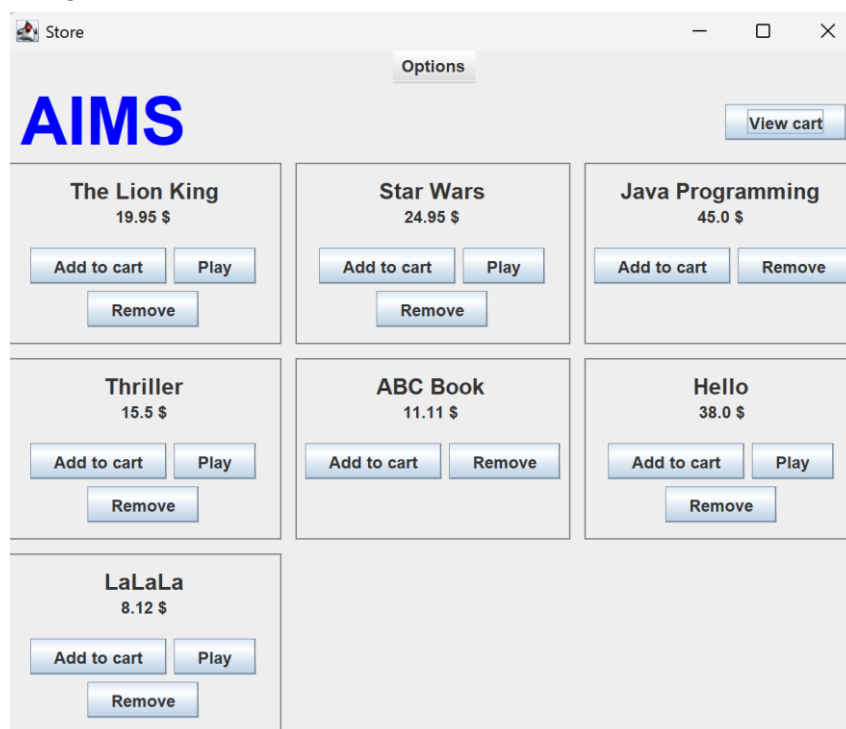
View cart

The Lion King 19.95 \$ Add to cart Play Remove	Star Wars 24.95 \$ Add to cart Play Remove	Java Programming 45.0 \$ Add to cart Remove
Thriller 15.5 \$ Add to cart Play Remove	ABC Book 11.11 \$ Add to cart Remove	Hello 38.0 \$ Add to cart Play Remove
LaLaLa 8.12 \$ Add to cart Play Remove		

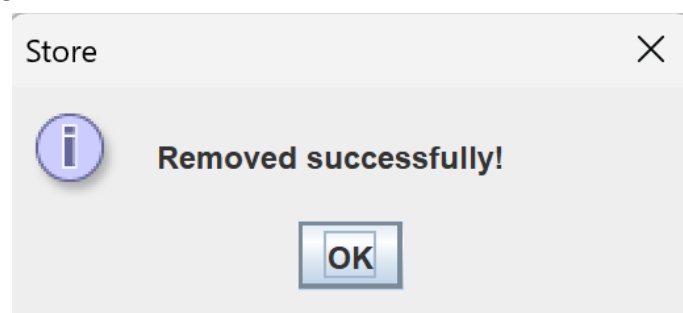
6. Test Remove 1 vài Item khỏi Store



- Nếu nhấn **No**

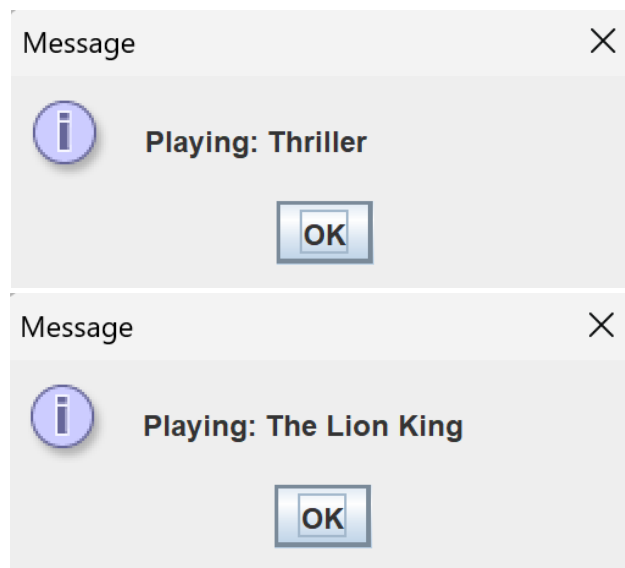


- Nếu nhấn **Yes**

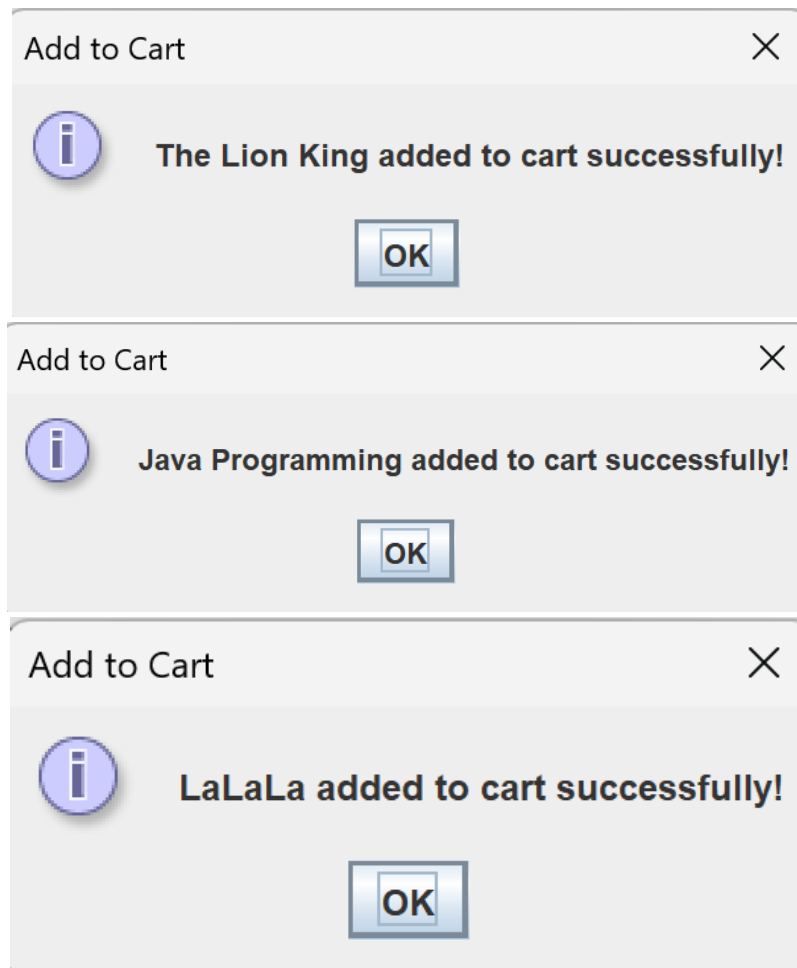




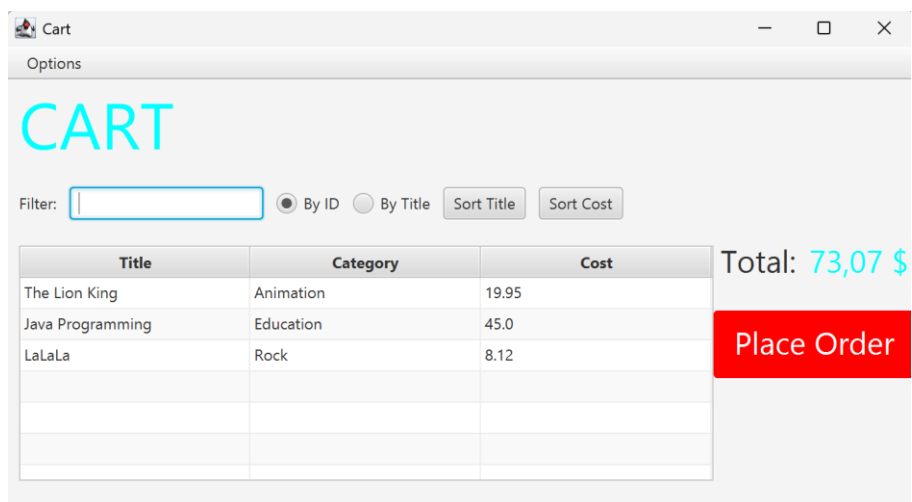
7. Play 1 vài Item DVD/CD trong Store



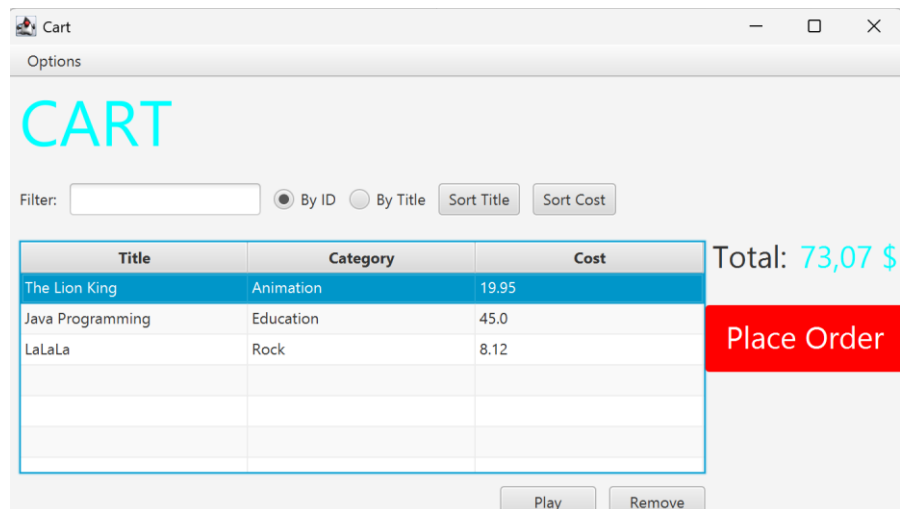
8. Add to Cart 1 vài cái



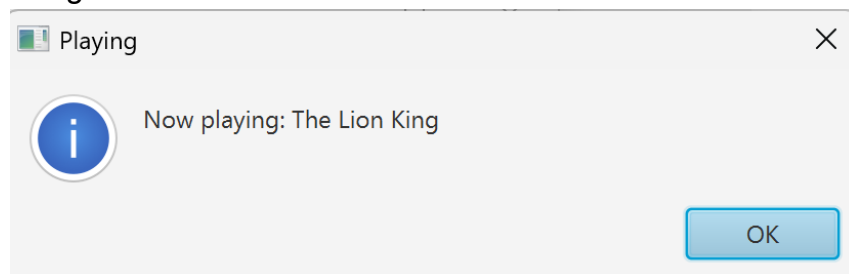
9. View Cart



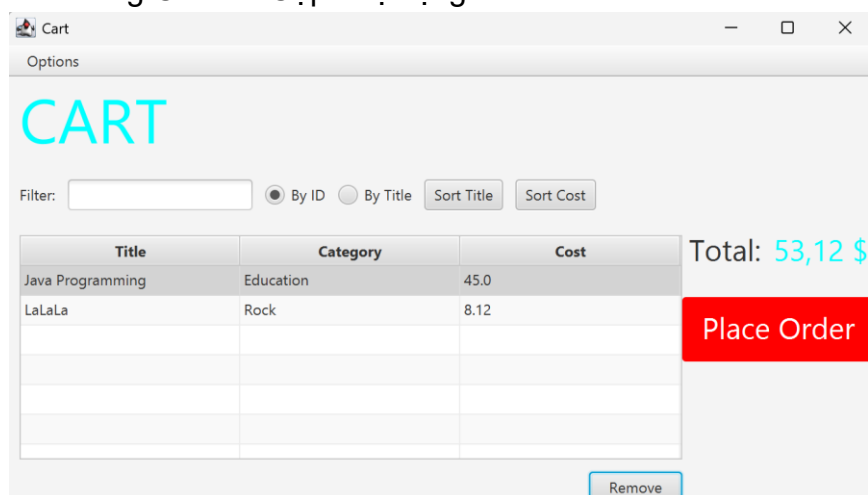
- Pick DVD/CD:



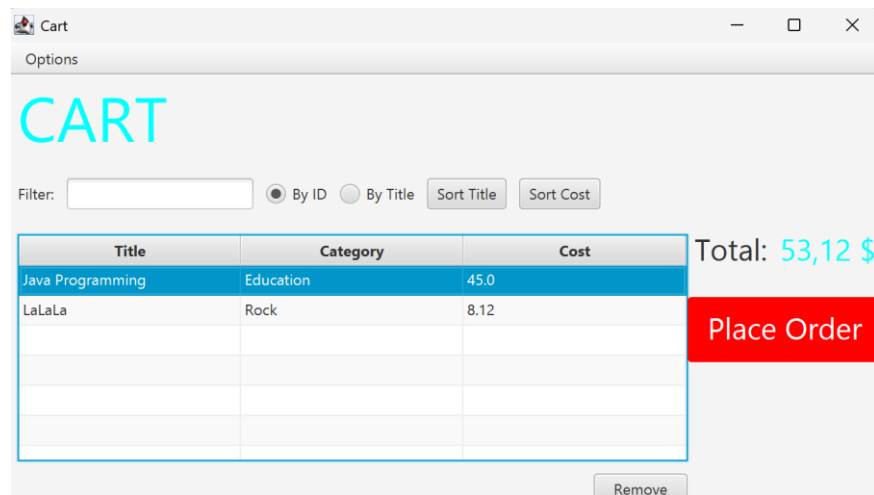
+ Play trong Cart



+ Remove trong Cart => Cập nhật lại giá tiền

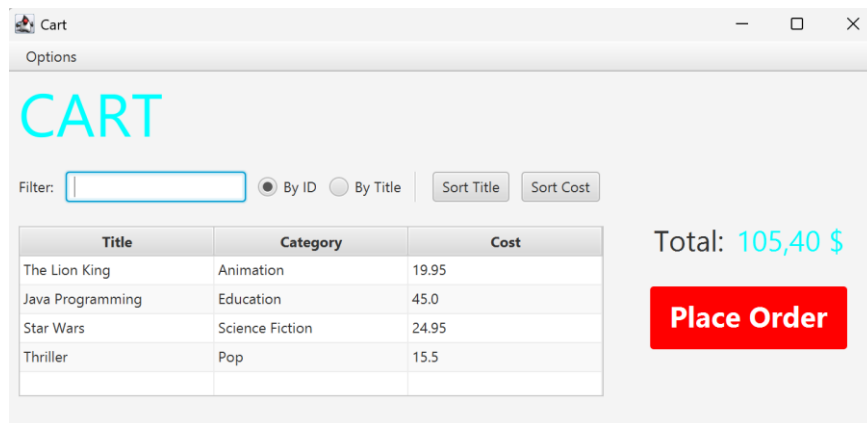


• Pick Book => KHÔNG CÓ PLAY

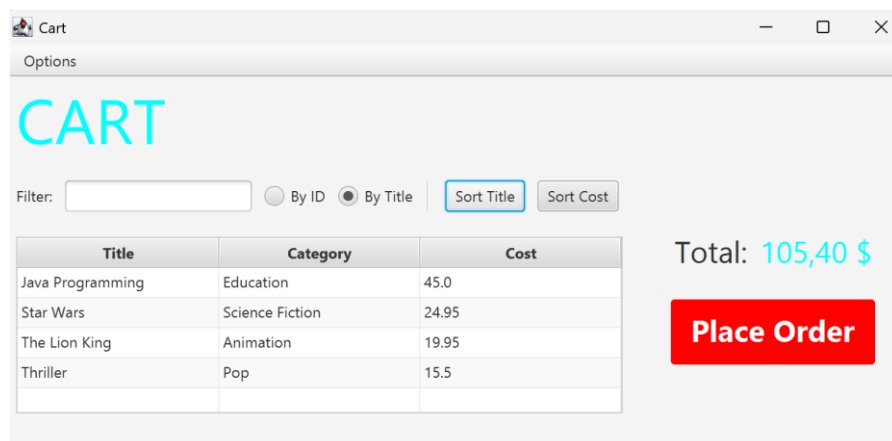


10. Add thêm to Cart để Test Sort

- Ban đầu:



- Sort Title



- Sort Cost

CART

Filter: ☐ By ID ☒ By Title

Title	Category	Cost
Thriller	Pop	15.5
The Lion King	Animation	19.95
Star Wars	Science Fiction	24.95
Java Programming	Education	45.0

Total: 105,40 \$

Place Order

11. Test Search by Filter

- Search by ID

Cho phép người dùng tìm kiếm sản phẩm trong giỏ hàng theo ID. Khi chọn chế độ “By ID”, hệ thống sẽ so sánh giá trị nhập vào với id của từng sản phẩm và hiển thị các kết quả phù hợp.

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
No content in table		

Total: 105,40 \$

Place Order

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
Star Wars	Science Fiction	24.95

Total: 105,40 \$

Place Order

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
Java Programming	Education	45.0

Total: 105,40 \$

Place Order

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
Thriller	Pop	15.5

Total: 105,40 \$

Place Order

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
The Lion King	Animation	19.95

Total: 105,40 \$

Place Order

CART

Filter: ☒ By ID ☐ By Title

Title	Category	Cost
No content in table		

Total: 105,40 \$

Place Order

- Search by Title

The first screenshot shows the filter set to 'e'. The table contains two items:

Title	Category	Cost
Java Programming	Education	45.0
Star Wars	Science Fiction	24.95

The second screenshot shows the filter set to 'e'. The table contains two items:

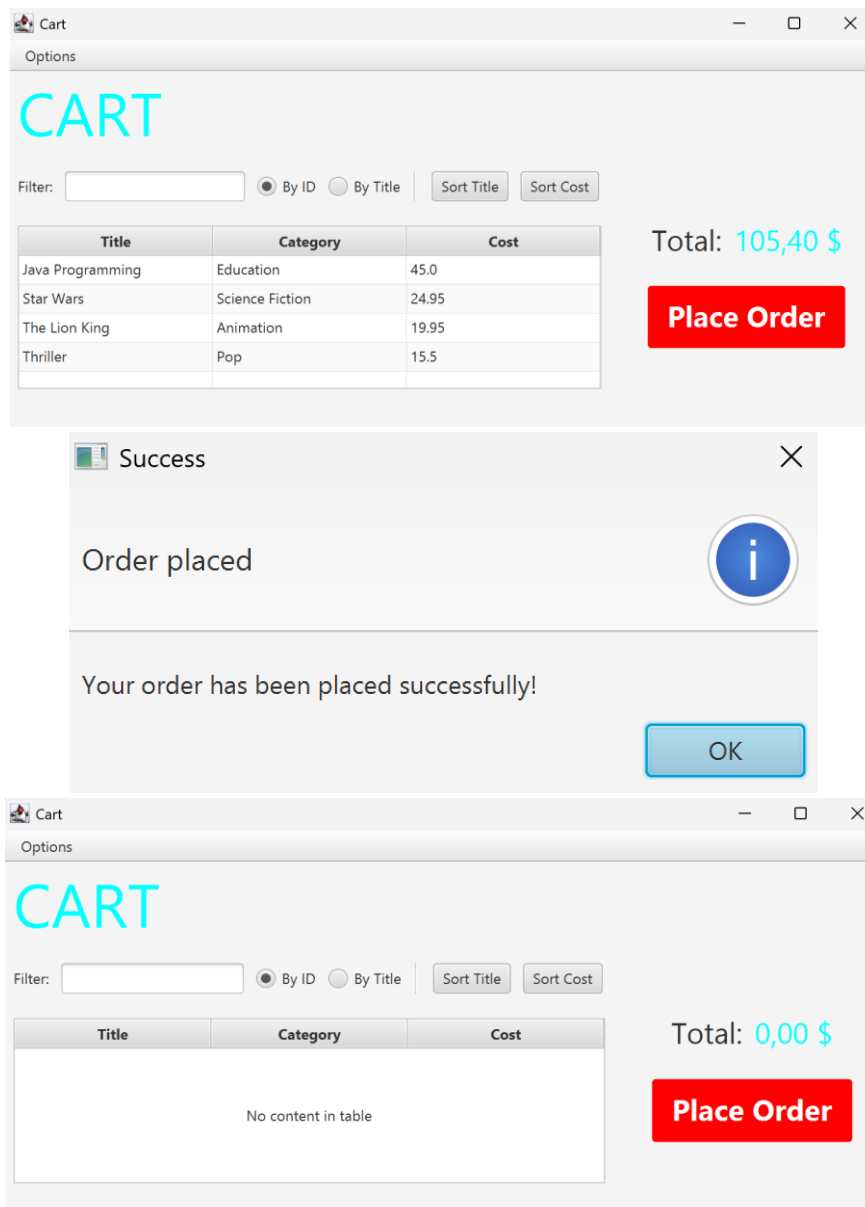
Title	Category	Cost
The Lion King	Animation	19.95
Thriller	Pop	15.5

The third screenshot shows the filter set to 'the'. The table contains one item:

Title	Category	Cost
The Lion King	Animation	19.95

12. Place Order

- Trường hợp trong Cart có Items



- Trường hợp Cart trống

